

# MetaMap: Facilitating Mapping Reuse and Lifecycle Documentation

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**Abstract**— In this research paper, we present ‘MetaMap’, a mapping annotation tool designed to serve an essential role in data and semantic mapping. MetaMap integrates key features for mapping annotation, incorporating metadata that meticulously captures the mapping lifecycle, with the hope of facilitating mapping reuse and mapping maintenance in subsequent mapping development phases. The tool's functionality unfolds in three core phases. Initially, users upload mappings and extract metadata. Then, users have the option to enrich their mappings with additional metadata fields, ensuring a comprehensive documentation from the lifecycle process of the associated mapping. Lastly, MetaMap empowers users with the flexibility to represent this metadata in various formats, such as named graphs or RDF-star.

**Keywords**— Mapping, Alignment, Metadata Model, Knowledge Representation, Annotation.

## I. INTRODUCTION

Declarative mappings are known as a valuable solution to enhance interoperability on the internet, where challenges stem from semantic heterogeneity and diverse data [1]. Heterogeneity and diversity is immediately evident when handling various datasets, ontologies, and publishing data online [2]. Declarative mappings offer a useful method for specifying the semantic relationships between different data models, addressing the intricacies of information exchange [1]. In the context of linked data and the semantic web, declarative mapping processes can be usefully categorized into three types: ontology mapping, interlinking, and uplift/download. Figure 1 provides an illustration of these mapping processes.

Each of the three mapping processes generates declarative mappings, albeit with distinct representations. Processed by engines, these mappings help with tasks like data transformation, alignment, and merging. Developing these mappings is complicated and requires expertise and time [3].

To save time and costs, it is important to reuse/repurpose previously made mappings and in general to manage and maintain the mappings given the evolution of datasets on either side of the mapping. Appropriate and detailed metadata associated with the mappings is clearly necessary to enable such development and management. While existing metadata associated with mappings is often limited and not easily queryable, prioritizing the mapping statement over the entire mapping development process, we address this common challenge in mapping metadata, as discussed in the related work section (Section II). This underscores the importance of a consistent documentation approach for mapping projects.



Figure 1 The Three categories of mappings

Our research focuses on a unified documentation model with detailed and accessible information with the aim of greatly benefiting knowledge engineers in their development, reuse and maintenance of mappings. The ultimate aim is to assist knowledge engineers in tasks such as finding, reusing, and evaluating the quality of mappings. As the foundation for the MetaMap tool, a mapping lifecycle has been formulated to accommodate all three types of mappings (presented earlier), along with a unified metadata model that comprehensively captures every aspect of mapping development—from its purpose and stakeholders to post-development, long-term management.

The proposed mapping lifecycle and metadata model was introduced in [4]. A survey experiment was conducted to collect feedback on the initial mapping metadata model from the semantic web and linked data community. The aim was to assess the relevance of the proposed metadata fields for two mapping-related tasks. Preliminary results, available in [5], indicate a generally positive response. In the experiment, most

of the proposed metadata fields were rated as relevant, and participants suggested additional useful metadata fields.

In the next section, we provide an overview of related work to position MetaMap within the broader context of existing research and tools in the field. Section III outlines the design and workflow of MetaMap, detailing its functions and user interfaces. The subsequent Section IV provides exploration of MetaMap's implementation, offering insights into technical considerations. In addition, Section V enriches the discussion with a practical case study. Finally, Section VI wraps up the paper by summarizing key points and discussing future research directions.

## II. RELATED WORK

Several studies in the literature have introduced diverse metadata models and methods for enriching mappings with metadata [6][3][7]. However, many of these studies predominantly focus on metadata associated solely with the final mapping artefact. In contrast, a few notable ones, including [6] and [3], take a more holistic approach by considering the entire lifecycle of mapping development, extending beyond the final artefact. The OM2R project is particularly notable for being the first to comprehensively capture different stages of the ontology mapping process through its metadata model [6]. OM2R provides a specialized vocabulary for describing mappings while maintaining a clear distinction from the mappings themselves. Alternatively, the proposal in [3] introduces a novel ontology mapping lifecycle by adopting a project-centric approach. The corresponding metadata model, built as an ontology in OWL 2, extends the provenance ontology PROV-O[8] to comprehensively cover all mapping process steps.

In the realm of metadata generation, [9] presents an approach to automatically generate provenance and metadata information for linked data, implemented in the RML toolchain. Dimou et al. have pointed out that the type of metadata supported by their approach can be extended to accommodate other vocabularies. In the meantime, the RMLEditor [10] records the activities of document creation and editing, as well as when and by whom they were committed. Throughout their work, they have focused primarily on how to generate and represent metadata fields rather than what to capture, and as a consequence have captured limited metadata about mappings. In [7], a community standard called Simple Standard for Ontological Mappings (SSSOM) has been proposed as a means of sharing semantic mappings between information entities. In SSSOM, mapping sets and mappings are specified with schemas, data models, and metadata elements [11]. SSSOM, however, is limited in that it focuses on ontology mappings and the capture of metadata that describes the statements of the mappings and the provenance of the mappings without considering the entire lifecycle of the mapping.

As a final observation, it should be noted that previous studies often design metadata models focused on particular mapping types, as seen in [3] and [7], where the proposed models are specifically crafted for ontology mappings. Overall,

there is a need for a unified metadata model applicable to any declarative mapping type, with extensibility options as needed. Furthermore, to enhance mapping discovery, reuse, and quality assessment, the unified metadata model should comprehensively capture all aspects of mapping development, ranging from the mapping purpose, and involved stakeholders to the long-term management post-development.

In comparison to the tools and studies in the state of the art, MetaMap is specifically designed for metadata enrichment for all three mapping types, considering the entire lifecycle of the mapping. Furthermore, in designing MetaMap the research explores different representations of metadata and how these metadata should be attached/associated with a mapping while maintaining the mapping structure.

## III. DESIGN AND USE

MetaMap is proposed as a web-based tool designed to equip knowledge engineers with an extensive and intricate metadata model, offering guidance throughout the process of annotation of declarative mappings. The tool's functionality extends the process to the reuse of mapping by empowering users to make informed decisions about the viability and appropriateness of reusing specific mappings.

The tool was developed in two distinct phases, each dedicated to creating, testing, and evaluating a specific component, with one component developed after the other. The initial components developed, namely the annotation and reuse components, were designed based on our research philosophy that emphasizes the role of metadata annotation in facilitating the reuse of mappings.

As discussed in Section I, the tool design began with the development of a mapping lifecycle capable of handling three different mapping types, accompanied by a comprehensive metadata model. Figure 2 illustrates the MetaMap solution design, highlighting four crucial aspects. Each of these aspects plays a vital role in guiding the tool's development. This section delves into a detailed discussion of each aspect, covering associated design decisions.

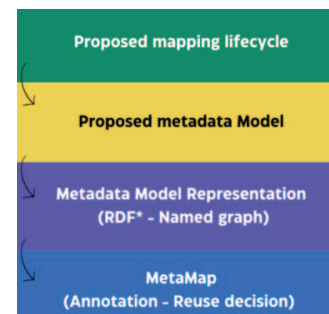


Figure 2 MetaMap solution design

### A. Mapping lifecycle

The proposed lifecycle is crafted for ease of use by engineers with a general software engineering background, following industry practices to facilitate the learning process. Comprising

six phases—Analysis, Mapping Design, Mapping Development, Mapping Testing, and Mapping Maintenance—the lifecycle is designed to be adaptable across the three declarative mapping types (ontology alignment, uplift/downlift, interlinking), with varying emphasis and application depending on the specific mapping type [4].

## B. Metadata model

Metadata fields were introduced for each stage of the lifecycle to capture and document the activities and decisions made during the corresponding phase of mapping development (46 fields in total) [4][5]. Additionally, a set of competency questions was formulated for each lifecycle stage to facilitate the validation of the proposed metadata model. Subsequently, the model underwent formalization into a specification. Within this specification, each field in the metadata model is distinctly defined by its associated vocabulary term, expected value type, and a comprehensive description of its purpose or significance. Table 1 illustrates how the specification describes the fields by presenting one example<sup>1</sup>.

*Table 1 the metadata model specification*

| Phase    | Field              | Vocabulary term                | Value   | purpose                                                                               |
|----------|--------------------|--------------------------------|---------|---------------------------------------------------------------------------------------|
| Analysis | Stakeholder's name | foaf:FirstName<br>foaf:surname | Literal | To provide a human readable name for the stakeholders involved in the mapping project |

Various RDF namespaces from well-known ontologies were employed to describe the metadata model. For instance, the FOAF vocabulary was utilized for capturing stakeholder information, encompassing their names and organizations. Additionally, the Dublin Core Metadata Initiative (DCMI) vocabulary was employed to detail aspects like the format, source, and creator of the input file. For metadata pertaining to the mapping itself, such as mapping requirements, domain, tools used, and mapping methods, custom namespace was created to encompass information about the mapping project. This approach ensures that metadata fields align with existing RDF vocabularies, enhancing interoperability and widespread comprehension.

## C. Metadata model representation

The representation and attachment of metadata to RDF-expressed mappings play a crucial role in storage efficiency, query performance, and usability [12]. Various RDF metadata approaches are found in the literature, such as standard reification, named graphs, singleton properties, and RDF-star. Standard RDF reification involves describing a triple using four statements, known as reification, with vocabulary terms like `rdf:Statement`, `rdf:predicate`, and `rdf:object`. This method has drawbacks, including the lack of formal semantics and a

significant increase in dataset size due to the need for four additional triples to represent one statement [12].

In an effort to overcome these drawbacks, [13] proposed the Singleton Property, offering formal semantics for statements about statements. This approach relies on predicates rather than subjects, amending the original predicate with a unique one to carry future metadata. Although it reduces the number of triples needed, it produces a large number of unique predicates [12].

Hartig & Thomson introduced RDFstar [14], extending the standard's syntax to allow nesting within the RDF graph. Using `<<` and `>>` to enclose referenced triples, this approach results in smaller documents, facilitating faster data exchange and shorter SPARQL queries [15].

Named graphs [16] represent another annotation approach, where an RDF graph is named with a URI. The identifier of the named graph in the RDF graph can be used to make simple statements about the entire named graph.

[12] provides a comparison of RDF reification approaches—standard reification, singleton property, and RDFstar—based on triples, storage size, and query execution time. Results show differences, with Singleton Property reducing triples by about 40%, RDF-star by almost 70%, and variations in storage size and query execution time.

Additional benchmarking reported in [15] tested approaches on a Wikidata subset. Although Singleton Properties were excluded due to the production of numerous unique predicates, RDF-star demonstrated favorable results. A recent study in the area [17] explores the suitability of various knowledge graph representations for different consumer scenarios: knowledge exploration, systematic querying, and graph completion. The research compares representations—Standard Reification, N-Ary Relationships, Wikidata qualifiers, Named Graphs, and RDF-star—through user studies, SPARQL querying analysis, and graph embedding model experiments. The study offers valuable insights into the importance of representation choices and their impact on knowledge graph consumption scenarios, contributing to the ongoing discussion on optimal knowledge graph deployment.

In consideration of benchmarking contributions and recent studies, RDF-star and named graphs were chosen as representations for the proposed metadata model.

## D. MetaMap

The initial phase of tool development outlined in this section primarily concentrated on crafting the annotation component. This component empowers users to upload mapping files and establish a metadata model associated with the mappings. Initially tailored to uplift mapping, the annotation component's functionality is expected to broaden progressively in future iterations, encompassing various mapping types and additional functionalities.

This section outlines the workflow, design, and output of the annotation component, specifically tailored for handling uplift

<sup>1</sup> <http://surl.li/oknaa>

mappings. As illustrated in Figure 3, the annotation process within MetaMap typically comprises four key steps:

- **Upload Mapping File:** Initiate the process by uploading the mapping file into the system. Initially, the system considers only uplift mapping, with plans to incorporate all other mapping types in the future.
- **Fill in Metadata Fields:** This step involves two subfunctions. The first subfunction detects metadata from the mapping file, while the second requires users to provide additional metadata details.
- **Annotate Mapping:** This step involves selecting a mapping representation, choosing between RDF-star and named graphs, and annotating the mapping accordingly.
- **Export result:** After a review process, the annotated content can be viewed or downloaded to a result file.

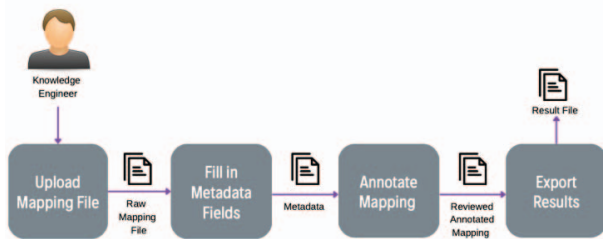


Figure 3 Typical workflow using MetaMap

The tool is publicly accessible [18], eliminating the requirement for user accounts. Users can effortlessly engage in mapping annotation and reuse directly through the web interface. Refer to Figure 4 for an illustration of the main website interface.



Figure 4 Home page of the MetaMap website

Following the previously outlined workflow, mapping annotation can be accomplished in four steps. The subsequent part of the section will demonstrate these steps with an example for annotating an RML mapping file.

In the example, the Attractions dataset, sourced from Data.gov.ie [19]—an Irish governmental open data portal—underwent uplift from CSV to RDF using the RMLEditor [10].

This process converts the data into RDF and generates mapping statements for subsequent annotation with MetaMap.

In the initial step, users are prompted to upload their mapping files. In this development phase, RML mapping files are accepted in Turtle format. With RDF Mapping Language (RML), heterogeneous data structures and serializations are mapped to RDF with customized rules. This mapping language is a superset of the W3C-standardized mapping language [R2RML], expanding its application and scope [20].

Upon uploading the mapping file, the tool automatically extracts recognized data, including the uploaded file name, mapped file names, and their formats—serving as the primary input for metadata, users proceed to another interface to add additional metadata fields. In this example a segment of the metadata model has been filled. As shown in Figure 5 These fields capture details related to the mapping development stage in the mapping lifecycle. Metadata includes the starting and ending dates of development, the tool utilized, as well as the mapping type and domain. After submitting metadata, users are prompted to review their entries and select their preferred presentation format. They have the option to choose between Named Graph, RDF-star, or both. Figure 5 provides a snapshot of the interface illustrating this step.

Figure 5 Select metadata representation

Upon a successful annotation process, users can either view or export the resulting file. Figure 6 displays the resulting file in the Named Graph format, while Figure 7 and 8 illustrates the result in RDF-Star. The RDF data model represents information in triples, comprising subject, predicate, and object. RDF triple stores accommodate multiple RDF graphs, allowing queries across different graphs concurrently. RDF datasets typically include a default graph (unnamed and potentially empty) and zero or more named graphs identified by IRIs, facilitating SPARQL queries for data retrieval [21]. To annotate the mapping using named graphs, the RML mapping statement file was treated as the default graph, followed by the creation of a new named graph (metadata graph) using SPARQL.



```

1  @prefix dcterms: <http://purl.org/dc/terms/> .
2  @prefix foaf: <http://xmlns.com/foaf/0.1/> .
3  @prefix ns1: <http://example.com/mappings/> .
4
5  ns1:subject ns1:endDate "2023-12-05" ;
6      ns1:format "Named Graph" ;
7      ns1:mappingDomain "Tourism" ;
8      ns1:mappingMethod "Automatic" ;
9      ns1:mappingType "Uplift Mapping" ;
10     ns1:startDate "2023-12-01" ;
11     ns1:toolUsed "RMLEditor" ;
12     dcterms:creator "Data.gov.ie" ;
13     dcterms:source "Data.gov.ie" ;
14     foaf:familyName "Alzahrani" ;
15     foaf:givenName "Sarah" ;
16     foaf:organization "TCD" .

```

Figure 6 Metadata Graph

For enriching the mapping with metadata using RDF-star, a detailed examination of the RML statement structure was necessary. In Figure 7, lines 2 to 3, metadata describing the input file was seamlessly integrated with the logical source triple, adding dcterms:creator and dcterms:source to the entire logical source triple. Subsequently, in Figure 8, lines 2 to 11, the remaining metadata fields were incorporated with the complete TriplesMap triple.

```

1  <<http://example.com#LogicalSource2>
2      dcterms:creator "Data.gov.ie";
3      dcterms:source "Data.gov.ie">>
4      rml:referenceFormulation ql:CSV;
5      rml:source "Attraction.csv";
6      a rml:LogicalSource.

```

Figure 7 logical source triple

```

1  <<http://example.com#TriplesMap0>
2      ns1:endDate "2023-12-05";
3      ns1:format "Named Graph";
4      ns1:mappingDomain "Tourism";
5      ns1:mappingMethod "Automatic";
6      ns1:mappingType "Uplift Mapping";
7      ns1:startDate "2023-12-01";
8      ns1:toolUsed "RMLEditor";
9      foaf:familyName "Alzahrani";
10     foaf:givenName "Sarah";
11     foaf:organization "TCD">>
12     rml:logicalSource <<http://example.com#LogicalSource2>>;
13     a rr:TriplesMap;
14     rr:predicateObjectMap ?pom;
15     rr:subjectMap ?sm.

```

Figure 8 TriplesMap triple

#### IV. IMPLEMENTATION

MetaMap, was developed using Python[22], SPARQL[23], SPARQL-Star [14], and RML[20]. Python, powered by the Flask library [24], played a central role in creating the web application and executing SPARQL queries through RDF libraries [25] for mapping annotation and graph creation. The user-friendly graphical interface was crafted using HyperText Markup Language (HTML) [26] and Cascading Style Sheets (CSS) [27]. In the initial stage of development, MetaMap supports declarative mappings of the

uplift type (specifically RML initially). The initial choice focus on using RML for transforming data into RDF was influenced by the necessity to accommodate various dataset formats. The code is available at the GitHub repository <sup>2</sup> which also contains a live demo website. In the second phase of development, the tool will expand its support to include the other types of mappings of interest in this research, that is interlinking and ontologies alignment.

#### V. FUTURE WORK AND CONCLUSION

Our initial research focused on the development of a mapping lifecycle and a foundational unified metadata model that will allow for the annotation and reuse of three major types of declarative mappings. The initial research laid the groundwork for the development of MetaMap—an innovative software application designed for annotating and reusing mappings. As detailed in Section III, MetaMap comprises two fundamental components that will be systematically developed in two distinct phases. The initial phase focuses on the creation of the annotation component, which is expounded upon in this paper. Subsequently, in a subsequent phase as part of future work, the development will progress to include a reuse decision support component, as well as support for ontology and interlinking types of mappings. In the near term, the usability and functionality of the current tool will undergo comprehensive evaluation by users, ensuring a thorough assessment of its practical application and effectiveness. Taking a user-centric evaluation will provide valuable insights into the tool's performance, informing potential refinements and optimizations in subsequent iterations.

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